Cloud Infrastructure 2

Storage Server (NAS)
Technical Documentation

Lycée Guillaume Kroll (LGK) BTS Cloud Computing – Semester 2 (2025–2026)

Team 6

Andrea Girotto · Guilherme Videira Marques

Server: **MR3S04**

UNRAID

Table of Contents

Table of Figures

1 Introduction

1.1 Project Context

This document presents the work carried out by **Team 6** during the **CLOIF2 – Cloud Infrastructure 2** module at Lycée Guillaume Kroll (LGK), as part of the BTS Cloud Computing Year 1 programme (2025–2026). The goal of the project was to stage, configure, and document a rack-mounted storage server (NAS) from bare metal to a fully operational, multi-user file-sharing and block-storage service.

The server assigned to Team 6 is identified as **MR3S04** – an Intel SR2600URBRPR 2U rack server. The chosen NAS operating system is **UnRAID** by Lime Technology.

1.2 Final System Overview

The complete installation provides the following services:

- **Hardware RAID 5** array managed by an Areca ARC-1880 controller, using 3 × SAS disks (280 GB usable), named areca_raid5
- **Software RAID 1** pool (Btrfs mirror) using 2 × native SATA disks (160 GB each), named software_raid1
- **File shares** accessible over SMB from Windows, Linux, and macOS: Photos, Media, Documents, Backup
- **Per-user access control** with three user accounts (gui, andrea, marios) and differentiated permissions
- **iSCSI block storage** target exposed via community plugin, tested and verified from a Windows initiator
- **Remote management** via Intel RMM3 (out-of-band) and the UnRAID web interface

2 Part 1 – Information & Research

2.1 NAS Research

2.1.1 What is a NAS?

A **NAS (Network Attached Storage)** is a dedicated storage device connected to a network. Unlike a workstation's local disk, a NAS is accessible to all devices on the network simultaneously, allowing multiple users and systems to read and write files from a central location without needing physical access to the hardware.

2.1.2 What is a NAS Used For?

A NAS provides centralised file storage for multiple users and devices. Common use cases include:

- **Backups** – centralised backup target for workstations, laptops, and mobile devices
- **Media storage** – shared photo, video, and music libraries accessible from any device on the network
- **Document management** – shared folders for teams or families with per-user access control
- **Virtualisation storage** – storage volumes for hypervisors via iSCSI or NFS
- **Surveillance** – storing footage from IP cameras

2.1.3 DAS, NAS, and SAN – Comparison

Type	Full Name	Description
DAS	Direct Attached Storage	Storage connected directly to one computer (e.g. via SATA or USB). Fast and simple, but only accessible to the local machine.
NAS	Network Attached Storage	File-level storage shared over a standard IP network. Multiple users and devices can connect simultaneously over SMB, NFS, or similar protocols.
SAN	Storage Area Network	Block-level storage over a dedicated high-speed network (e.g. Fibre Channel or iSCSI). Used in enterprise environments for servers and virtual machines that need raw disk performance.

2.1.4 Why Compute and Storage Are Separated

In professional environments, compute resources (CPU, RAM) and storage are placed on separate systems. This separation provides:

- **Performance** – storage I/O does not compete with processing workloads
- **Scalability** – storage capacity can be expanded independently of compute nodes
- **Reliability** – a storage failure does not bring down compute servers, and vice versa
- **Manageability** – storage can be backed up, replicated, and secured as an independent system

This is the model followed in this project: server MR3S04 acts as a dedicated NAS, serving storage to clients over the network rather than processing applications locally.

2.2 Hardware Analysis

2.2.1 Server Form Factor and Dimensions

The server is an **Intel SR2600URBRPR**, a standard **19-inch rack-mount, 2U** chassis.

The **BRPR** suffix identifies this as the **passive / non-redundant fan variant**: it ships with a passive PCIe riser (no active SAS/SATA midplane) and three fixed non-redundant system fans. The alternative variant, SR2600URLX, is the active version with a redundant fan module and an active SAS/SATA midplane.

Field	Value
Product code	SR2600URBRPR
Model	Intel Server System SR2600URBRPR
Server board	Intel Server Board S5520UR
Serial No.	BZDT10400131
TA No.	E86263 - 008

Rack unit reference:

Measurement	Inches	cm	mm
Standard rack width (rail-to-rail)	17.75 in	45.09 cm	450.85 mm
1U height	1.75 in	4.445 cm	44.45 mm
2U height	**3.50 in**	**8.89 cm**	**88.9 mm**

Server chassis dimensions (H × W × D):

Measurement	Inches	cm	mm
Standard rack mounting width	19.00 in	48.26 cm	482.6 mm
Server height (2U)	3.50 in	8.89 cm	88.9 mm
Chassis height	3.44 in	8.74 cm	87.4 mm
Chassis width	16.93 in	43.00 cm	430.0 mm
Chassis depth	27.75 in	70.48 cm	704.8 mm
Maximum chassis weight	—	—	29.5 kg (65 lbs)

Note: The metal chassis body is narrower than 19 inches. The 19-inch figure refers to the rack mounting standard, measured across the ears and rails — not the chassis body alone.

2.2.2 External Ports and Connectors

Port / Connector	Count	Purpose
VGA (rear)	1	Connect a local monitor for initial setup and troubleshooting
VGA (front)	1	Front video port for direct local access; cannot be used simultaneously with rear VGA
USB 2.0 (rear)	4	Keyboard, mouse, USB installer, maintenance tools
USB 2.0 (front)	1	Front-panel USB for local access and diagnostics
Serial port (RJ-45, rear)	1	Serial console access (Port A)
LAN (mainboard NIC)	2	Dual 10/100/1000 Mb/s Ethernet via Intel 82575 PHY; standard network connection for the server
RMM (remote management)	1	Out-of-band access via Intel RMM3: remote BIOS, KVM console, power control (IPMI 2.0 compliant)
RAID controller LAN	1	Dedicated port for the Areca ARC-1880 web management interface

2.2.3 Internal Hardware Components

Component	Installed / Observed	Purpose
Mainboard	Intel Server Board S5520UR	Connects and controls all hardware components; hosts the Intel 5520 chipset I/O Hub and Intel 82801Jx I/O Controller Hub
CPUs	2 × Intel Xeon X5650 @ 2.67 GHz	Server processing power (6-core each, 12 cores total); FC-LGA 1366 Socket B, 95 W TDP, 6.4 GT/s Intel QPI
CPU heatsinks	2 (one per CPU)	Thermal dissipation
RAM slots	12 slots total (4 used per CPU side)	DDR3 ECC RDIMM or UDIMM, 800/1066/1333 MHz; 12 slots across 6 channels (3 per CPU); 8 × 2 GB installed = 16 GB total
Drive bays	5 × 3.5-inch hot-swap SATA/SAS bays + 1 flex bay	Front-access hot-swap bays for OS disks and data; flex bay supports a 6th 3.5" HDD, tape drive, or 2 × fixed 2.5" drives. Only 2 SATA bays used in this project.
Drive backplane	Hot-swap SAS/SATA backplane	Connects front drive bays to the mainboard SATA ports
PCIe expansion	Passive 2U riser (3 × full-height PCIe slots)	Hosts the Areca ARC-1880 in slot 1; 2 slots remain free. Passive riser = no active SAS midplane.
Power supply bays	2 × slidable PSU slots	Up to two 750 W modules; supports redundant operation (both PSUs installed)
Cooling fans	3 fixed non-redundant fans	Active airflow from front to rear; non-redundant on BRPR variant (no hot-swap replacement)
Airflow shroud	Processor air duct + large air baffle	Directs airflow over CPUs and memory; must be reinstalled after any maintenance
BMC	Integrated on S5520UR (Server Engines Pilot II)	Baseboard Management Controller; IPMI 2.0 compliant; powers the RMM3 out-of-band management

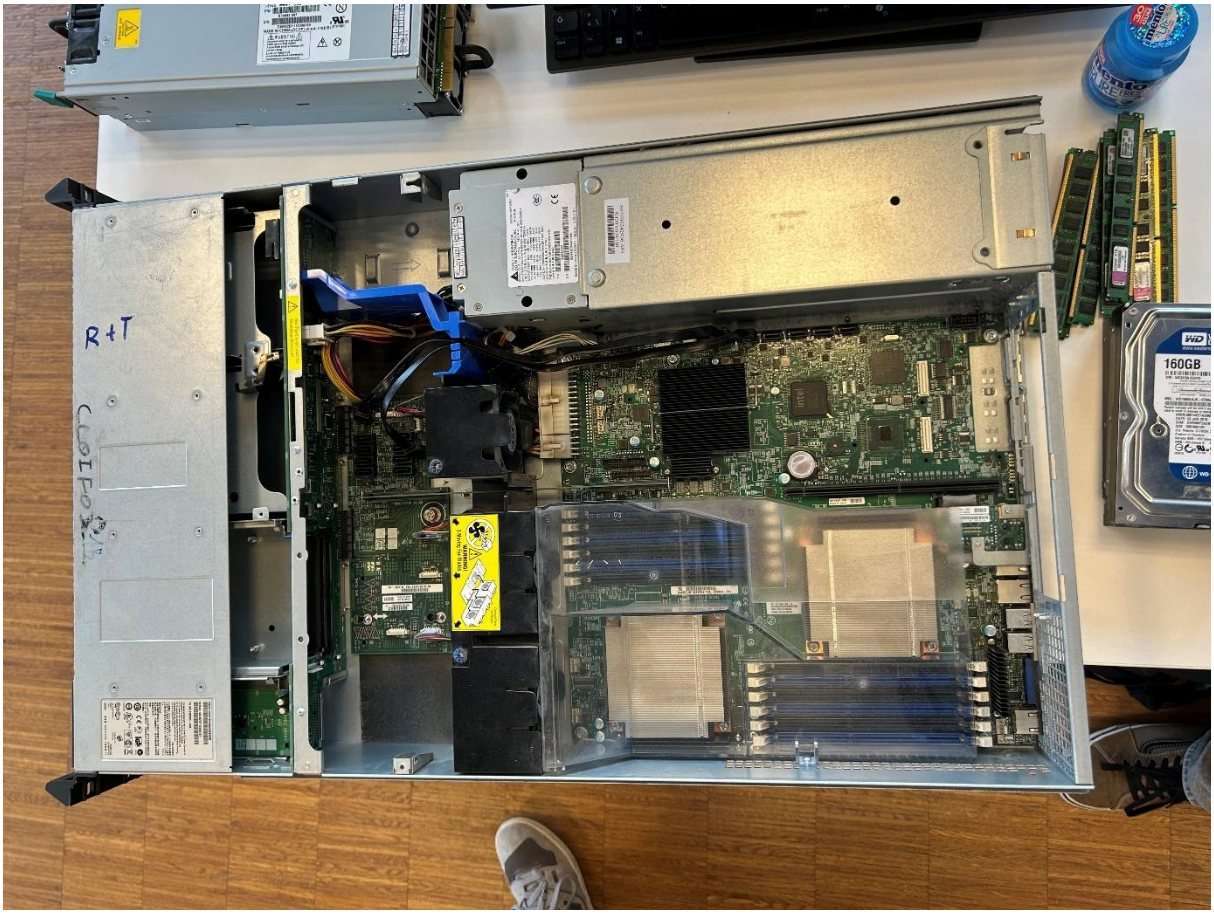


Figure 1: Open server chassis showing the internal layout and major hardware components of the Intel SR2600URBRPR server.

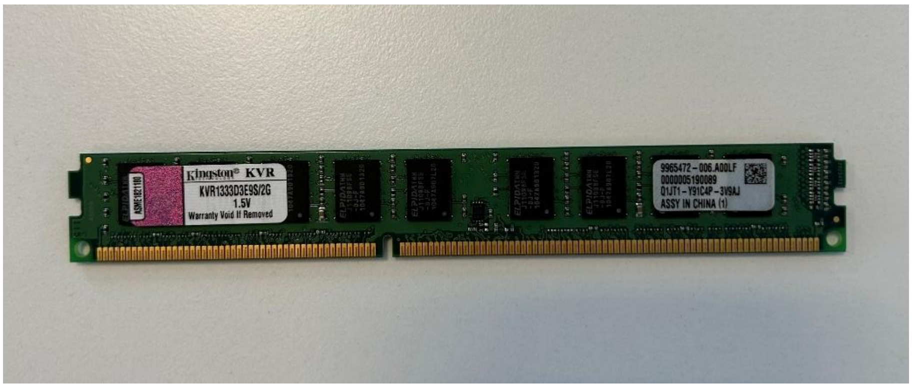


Figure 2: Installed Kingston DDR3 memory modules on both CPU sides of the server mainboard.

2.2.4 Internal Ports and Connectors

Port / Connector	Physical Location	Purpose
CPU sockets (2 × LGA 1366 Socket B)	Mainboard	Mount Intel Xeon processors (up to 95 W TDP each)
RAM slots (12 × DDR3 DIMM)	Mainboard, flanking both CPUs (6 per CPU)	Install ECC registered or unbuffered DDR3 memory; 3 channels per CPU
6 × SATA II connectors	Mainboard	Connect native SATA disks directly to the mainboard; 2 used (SATA_0, SATA_1) for the WD drives
SATA backplane	Front drive bay assembly	Routes SATA signals from mainboard to the 5 hot-swap front bays
PCIe slots (3 × full-height)	Passive 2U riser card	PCIe expansion; slot 1 used by Areca ARC-1880; 2 slots free
SAS ports (on Areca card)	Areca ARC-1880 PCIe card	Connect SAS disks to the hardware RAID controller; 3 used for Cheetah drives
RMM3 header	Mainboard (dedicated connector)	Plugs into Intel RMM3 add-in module for out-of-band BMC/IPMI access
USB 2.0 header (internal)	Mainboard	Supports 2 additional USB ports internally (e.g. for UnRAID USB key on rear port)
Fan headers	Mainboard	Power and speed control for the three fixed cooling fans
24-pin ATX + 8-pin CPU power	Mainboard	Main power delivery from PSU to mainboard and CPUs
PSU power connectors	Drives, backplane	Power distribution from PSUs to drives and peripheral components

3 Part 2 – Server Staging

3.1 Peripherals Connected

Before starting hardware installation, the following peripherals were connected for local access during the staging session:

- Monitor via **VGA**
- Keyboard via **USB**
- Mouse via **USB**

3.2 Hardware Installation

3.2.1 RAM

Parameter	Value
Module model	Kingston KVR1333D3S8/2G
Type	DDR3, 1333 MHz
Capacity per module	2 GB
Modules installed	8 (4 per CPU side)
Slots used per side	4 of 6
Total installed RAM	**16 GB**
Verified in BIOS	Yes

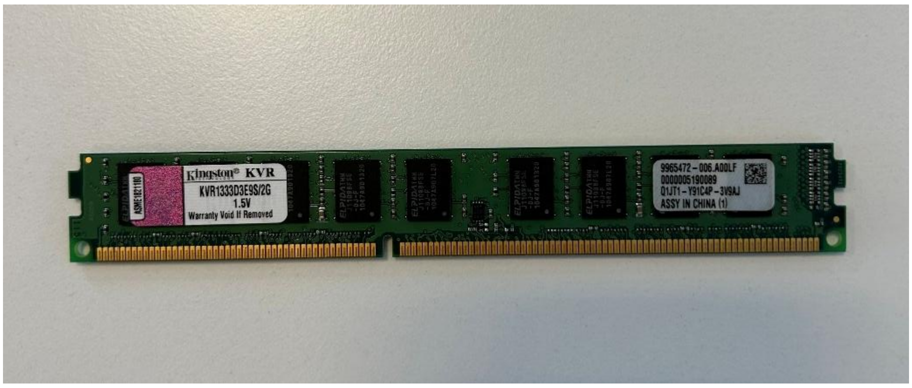


Figure 3: Eight Kingston KVR1333D3S8/2G memory modules installed, providing a total of 16 GB of system RAM.

3.2.2 Native SATA Disks

Two Western Digital Caviar Blue drives were installed on the server's native SATA ports. These two disks are later used to create the software RAID 1 pool.

Drive label	Model	Interface	Capacity	Connected to
DRIVE_0	WD1600AAJS-07M0A0	SATA	160 GB	SATA_0
DRIVE_1	WD1600AAJS-07M0A0	SATA	160 GB	SATA_1

Figure 4: Two Western Digital WD1600AAJS SATA disks installed on the native motherboard SATA ports.

3.2.3 RAID Controller and SAS Disks

The Areca ARC-1880 RAID controller was installed in the PCIe riser slot. Three Seagate Cheetah SAS disks were connected to the controller to form the hardware RAID 5 array.

Component	Value
RAID controller	Areca ARC-1880 (PCB label: ARC18801 VER:C)
Controller interface	PCIe Express
Installed in	PCIe riser slot
Backup battery	Removed — old battery caused continuous beeping; removed on teacher's recommendation. Controller operates normally without it.

Drive label	Model	Interface	Speed	Capacity	Connected to
DRIVE_2	Seagate ST3146356SS (Cheetah 15K.6)	SAS	15,000 RPM	146 GB	Areca SAS port 0
DRIVE_3	Seagate ST3146356SS (Cheetah 15K.6)	SAS	15,000 RPM	146 GB	Areca SAS port 1
DRIVE_4	Seagate ST3146356SS (Cheetah 15K.6)	SAS	15,000 RPM	146 GB	Areca SAS port 2

Figure 5: Three Seagate Cheetah 15K.6 SAS disks prepared for the hardware RAID 5 array.

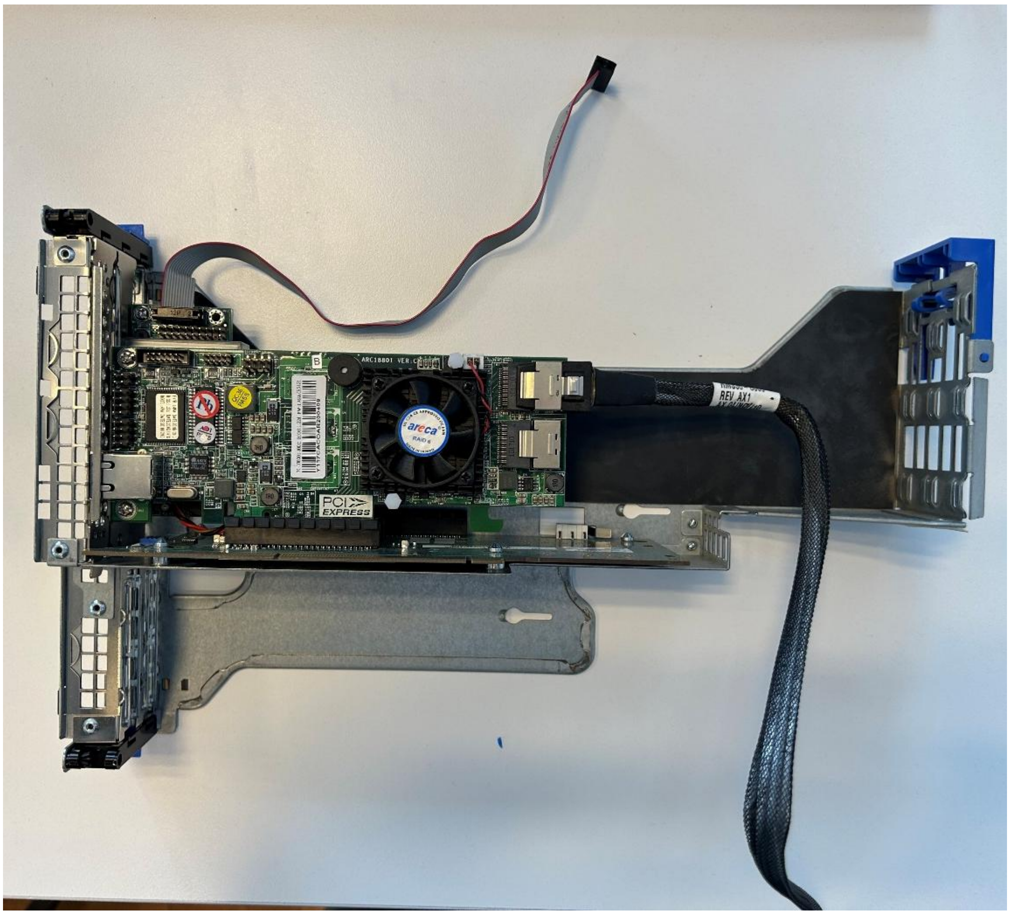


Figure 6: Areca ARC-1880 RAID controller installed in the PCIe riser during hardware staging.

3.3 Power Supply Testing

Parameter	Value
PSU model	Delta Electronics DPS-750PB A
Wattage	750 W each
Configuration	Redundant (2 × PSU)

Test procedure and results:

1. **PSU 1 removed** while the server was running → server continued operating without interruption.
2. **PSU 2 removed** while the server was running → server continued operating without interruption.

The redundant PSU configuration worked correctly. Either PSU is capable of powering the server alone, and the system fails over transparently with no visible interruption.

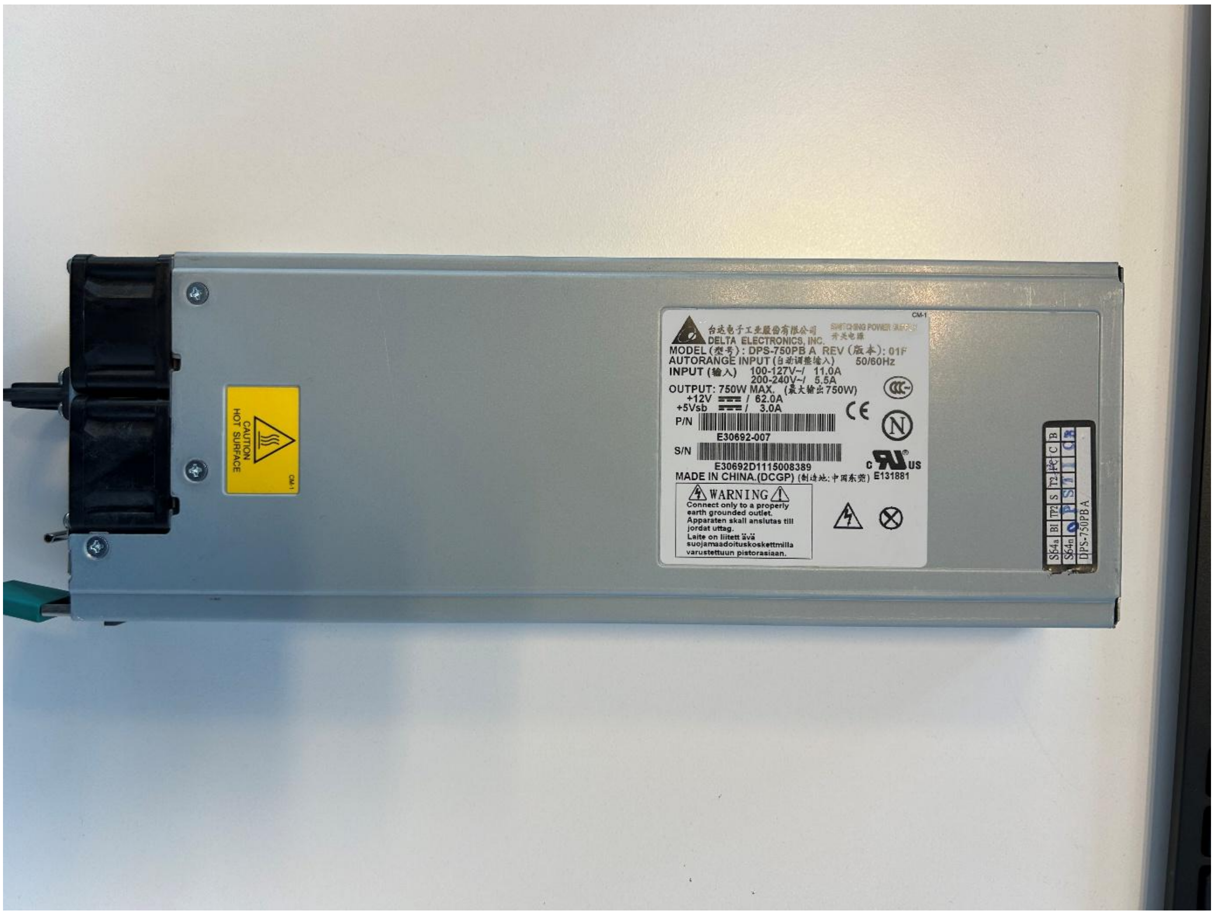


Figure 7: Redundant Delta 750 W power supply bays used for PSU failover testing.

3.4 BIOS Configuration

3.4.1 Hardware Detection

After all hardware was installed, the following was verified in the BIOS:

Component	Detected
RAM (16 GB)	✓ Yes
Native SATA disks (2 × WD1600AAJS)	✓ Yes
RAID controller (Areca ARC-1880)	✓ Yes
SAS disks (visible via controller)	✓ Yes

3.4.2 BIOS Settings

Setting	Value
RMM IP	192.168.0.42
USB Boot Priority	Enabled
EFI Optimised Boot	Disabled
Boot Option Retry	Disabled
Static Boot Ordering	Disabled

Boot order when no USB is connected:

Priority	Boot option
#1	SATA0 (WD1600AAJS)
#2	Internal EFI Shell
#3	IBA GE Slot 6100 (PXE / network)

When an UnRAID USB is connected, USB Boot Priority causes it to boot first.

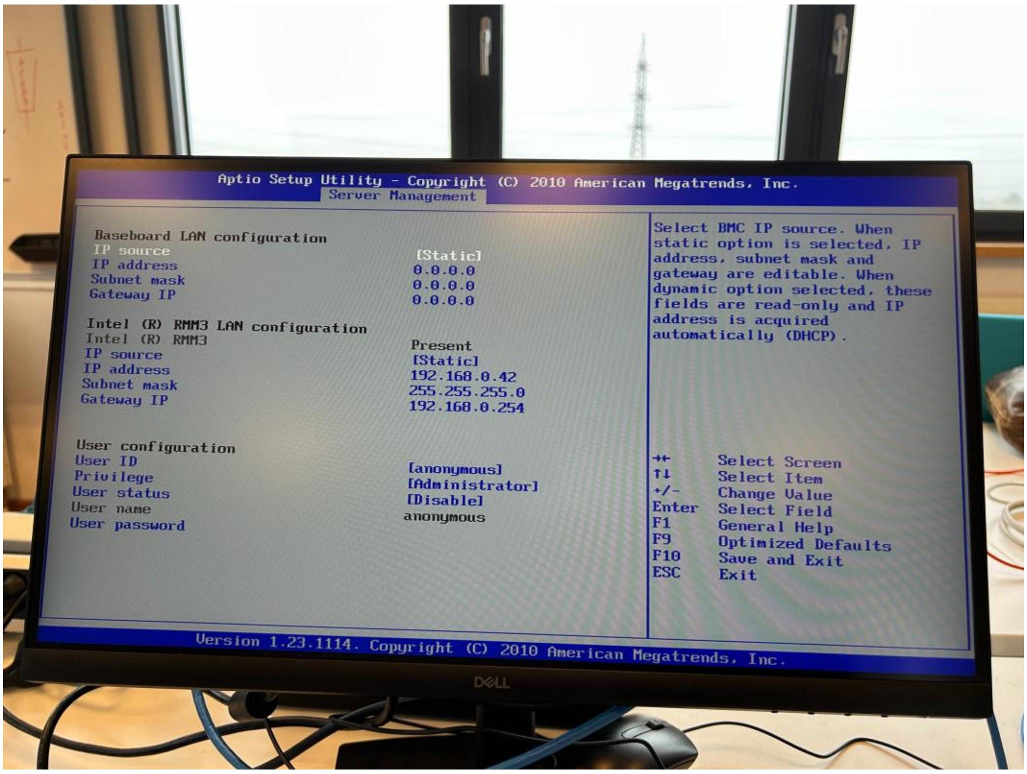


Figure 8: BIOS screen showing the configured Intel RMM management IP address `192.168.0.42`.

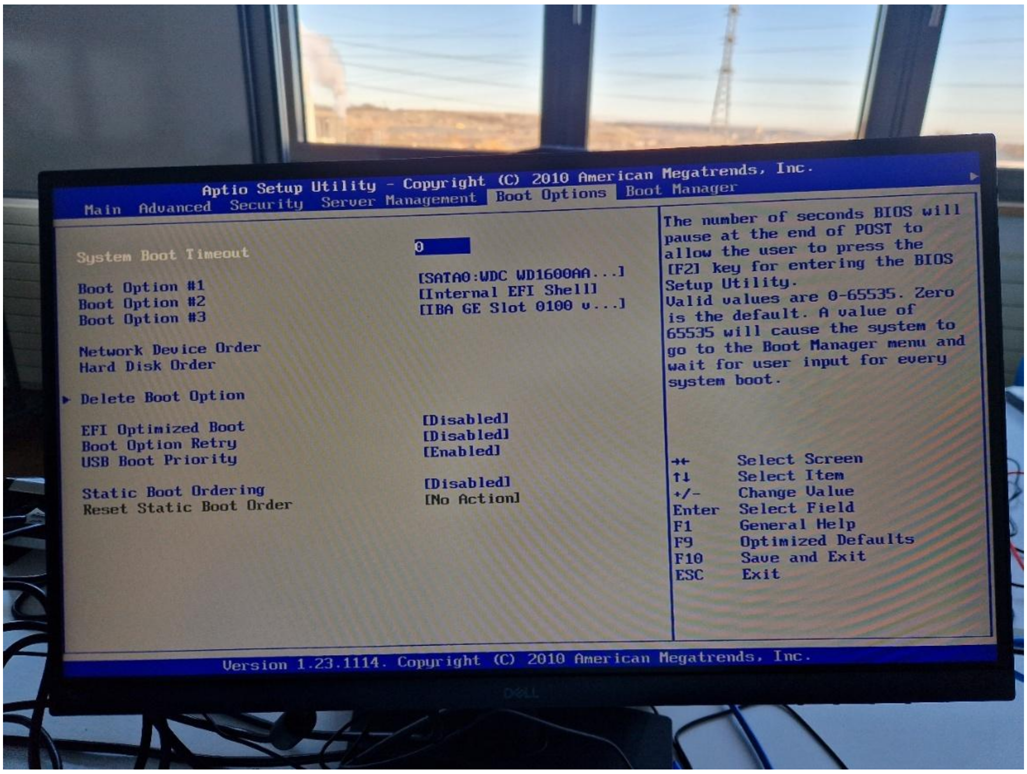


Figure 9: BIOS boot configuration showing USB boot priority and the default boot order.

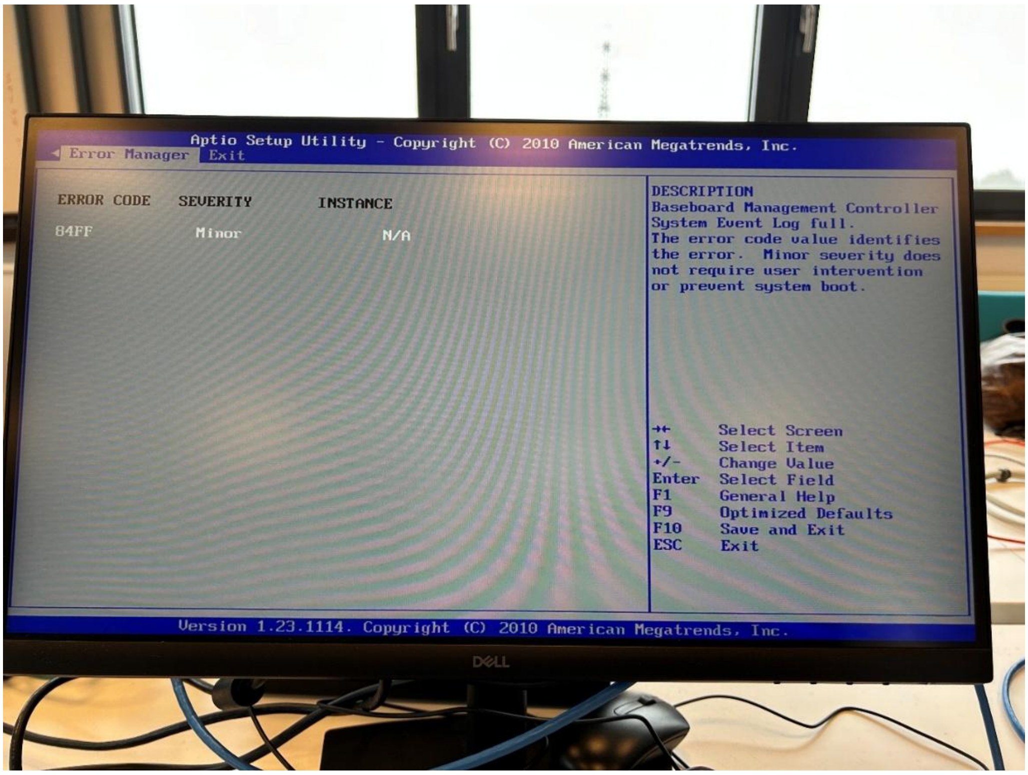


Figure 10: BIOS warning or error screen observed during staging; keep this caption generic until the exact message is confirmed.

3.4.3 RAID Controller IP Verification

The Areca ARC-1880 management IP `192.168.0.142` was verified in the BIOS and confirmed reachable from the lab network after connecting the controller's management port.

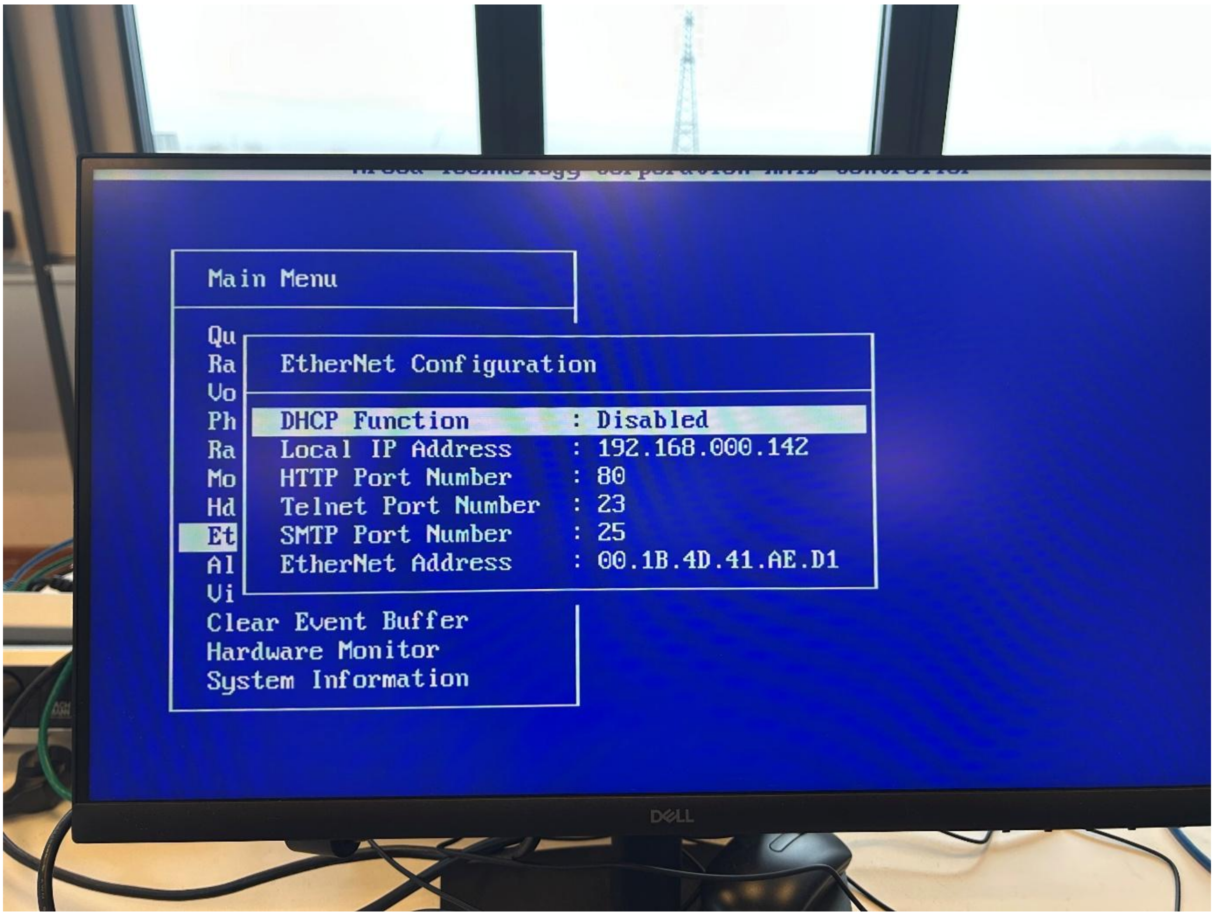


Figure 11: RAID controller BIOS screen confirming the Areca management IP address `192.168.0.142`.

4 Part 3 – RAID Controller Configuration

4.1 Network Connection

The Areca ARC-1880 has a dedicated Ethernet management port. This port was connected to the lab LAN with the pre-assigned IP address 192.168.0.142.

4.2 Web Interface Access

The controller's web management interface was accessed at:

http://192.168.0.142

Login credentials: username admin, password 0000.

4.3 RAID 5 Array Creation

The following steps were performed in the Areca web interface:

1. **Delete existing RAID set** — Raid Set Functions → Delete Raid Set
2. **Create new RAID set** — Raid Set Functions → Create Raid Set (select all 3 SAS disks)
3. **Create volume set** — Volume Set Functions → Create Volume Set (RAID level: 5, name: areca_raid5)
4. **Configure Ethernet** — System Controls → Ethernet Configuration (IP: 192.168.0.142, DHCP: disabled)

Parameter	Value
Volume name	areca_raid5
RAID level	RAID 5
Disks	3 × Seagate Cheetah 15K.6 ST3146356SS (146 GB each)
Usable capacity	**280.0 GB**
Volume state after creation	**Normal** (online immediately, fully operational)

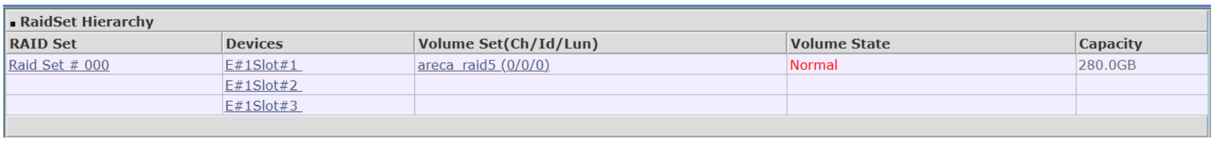


Figure 12: Areca web interface showing the RAID 5 set hierarchy and the normal status of `areca_raid5`.

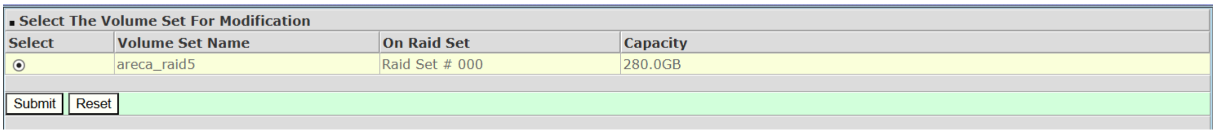
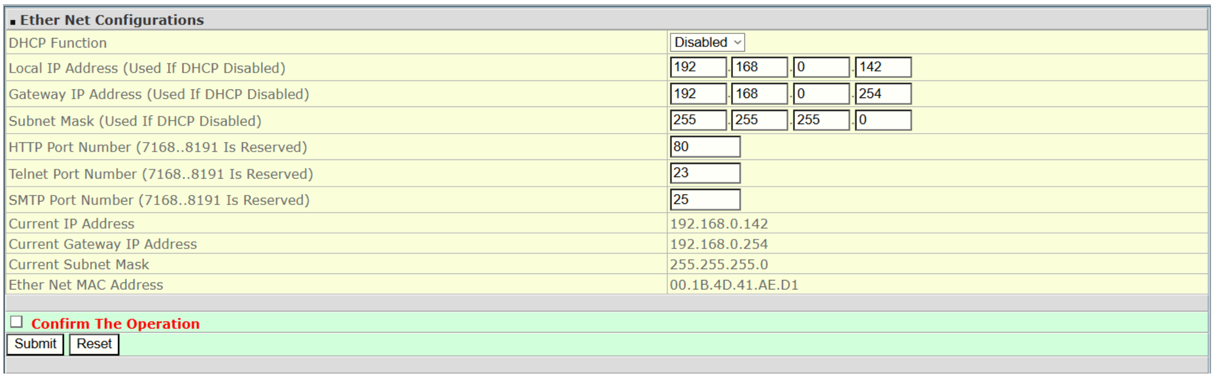


Figure 13: Areca volume set summary for `areca_raid5` on Raid Set `#000`, showing the created logical volume.



Ether Net Configurations	
DHCP Function	Disabled
Local IP Address (Used If DHCP Disabled)	192 . 168 . 0 . 142
Gateway IP Address (Used If DHCP Disabled)	192 . 168 . 0 . 254
Subnet Mask (Used If DHCP Disabled)	255 . 255 . 255 . 0
HTTP Port Number (7168..8191 Is Reserved)	80
Telnet Port Number (7168..8191 Is Reserved)	23
SMTP Port Number (7168..8191 Is Reserved)	25
Current IP Address	192.168.0.142
Current Gateway IP Address	192.168.0.254
Current Subnet Mask	255.255.255.0
Ether Net MAC Address	00.1B.4D.41.AE.D1

Confirm The Operation

Submit Reset

Figure 14: Areca Ethernet configuration with static IP `192.168.0.142` and DHCP disabled.

5 Part 4 – OS Installation

5.1 NAS Software – UnRAID

5.1.1 Overview

UnRAID is a Linux-based NAS operating system developed by **Lime Technology, Inc.** It is designed to run on standard PC or server hardware and is particularly known for its flexible approach to storage: disks of different capacities can be combined in the same array while still providing parity-based protection against disk failure.

Field	Value
Developer	Lime Technology, Inc.
Website	https://unraid.net
Latest release (at documentation date)	7.2.4 (released 2026-02-24)
Version used in our lab	Updated from 7.1.2 to 7.2.4
Licence type	Commercial (paid, perpetual)

5.1.2 Licensing

UnRAID requires a paid licence. Three tiers are available:

Tier	Price	Storage devices	Updates included
Starter	$49 (one-time)	Up to 6	1 year
Unleashed	$109 (one-time)	Unlimited	1 year
Lifetime	$249 (one-time)	Unlimited	Lifetime

5.1.3 Key Features

- Mixed-size disk support in a single array
- Parity-based protection (survives one disk failure in a standard array)
- Docker container hosting
- Virtual machine support
- Web-based administration interface
- Extensible plugin ecosystem (Community Apps)

5.1.4 Advantages and Disadvantages

Advantages	Disadvantages
Flexible disk expansion – disks of different sizes can coexist	Commercial licence required – not free or open source
User-friendly web management interface	Parity array does not stripe data, so throughput can be lower than traditional RAID 5
Strong app ecosystem: Docker containers, plugins, VMs	Starter and Unleashed include only 1 year of updates; Lifetime licence required for ongoing updates

5.2 USB Preparation

The teacher provided a **pre-configured, hardware-bound UnRAID USB stick** with a valid licence already installed. We were instructed to use it as-is, without performing a fresh OS installation.

Configuration reset procedure:

1. Opened the USB contents in File Explorer.
2. Navigated to the config folder on the USB.
3. Deleted all files inside config **except** the licence key file.
4. This preserved the valid hardware-bound licence while removing the previous team's configuration, providing a clean starting state.

5.3 Boot and Initial Configuration

5.3.1 Boot

USB Boot Priority was already enabled in the BIOS (confirmed in Part 2). The server was booted from the **rear USB port**, where the UnRAID stick was installed. (The licence is hardware-bound to that specific USB device.)

The boot menu entry order in /boot/syslinux/syslinux.cfg was adjusted by adding menu default to the GUI boot entry, so the graphical interface was selected automatically on startup.

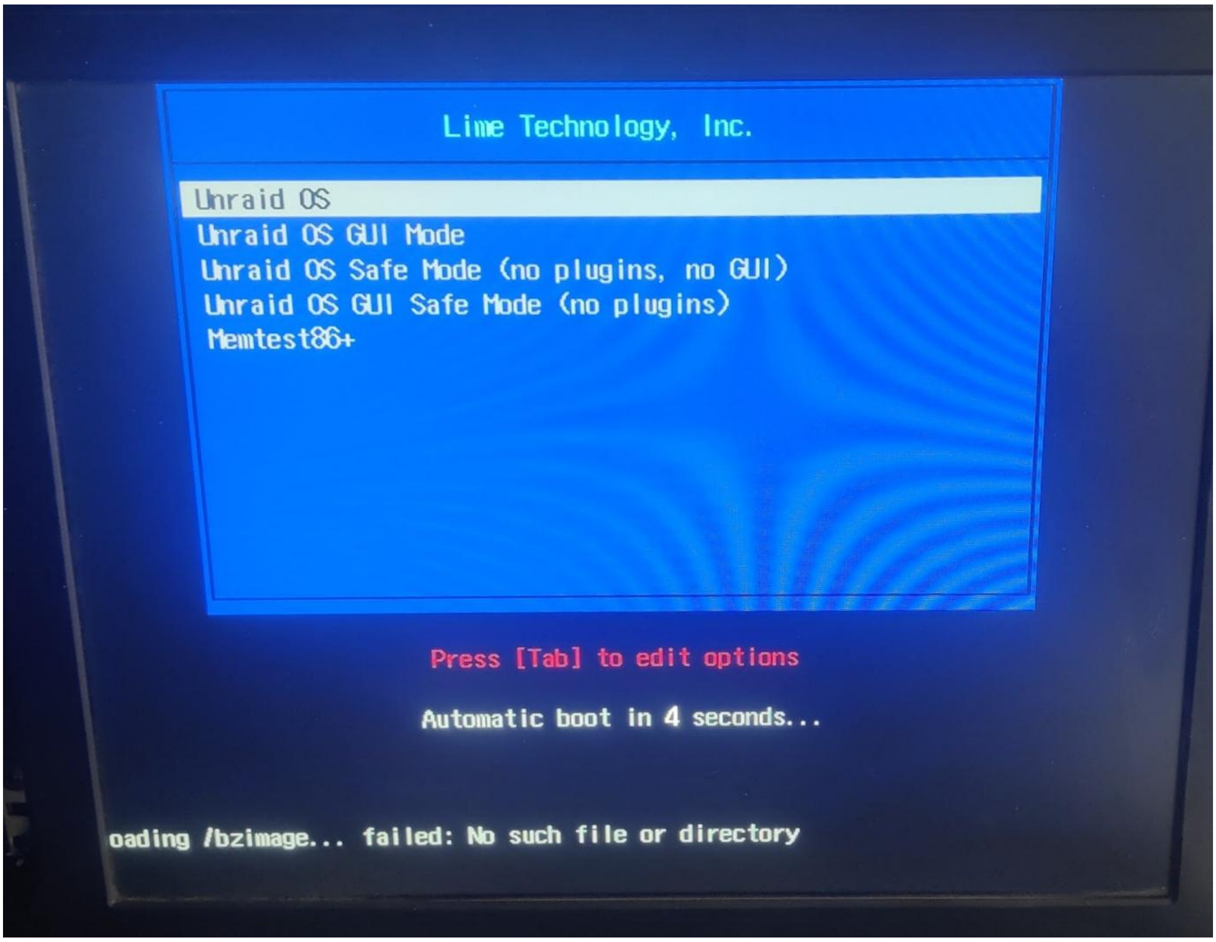


Figure 15: UnRAID boot manager with the GUI entry used to start the system from the licensed USB key.

5.3.2 Network and System Settings

After boot, the following configuration was applied through the UnRAID web interface:

Setting	Value
Username	`root`
Password	`test..123`
Hostname	`MR3S04`
IP assignment	`Static`
IPv4 address	`10.0.13.4`
Subnet mask	`255.255.0.0 (/16)`
Default gateway	`10.0.0.1`
Primary DNS	`10.0.0.1`
Secondary DNS	`1.1.1.1`

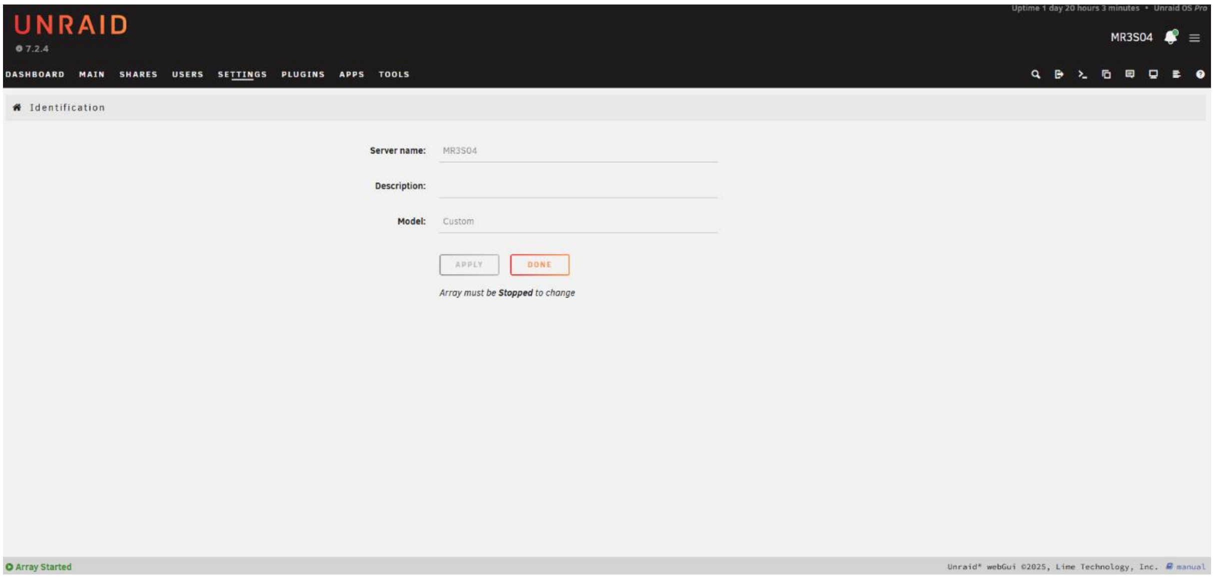


Figure 16: UnRAID identification settings showing the server hostname configured as `MR3S04`.

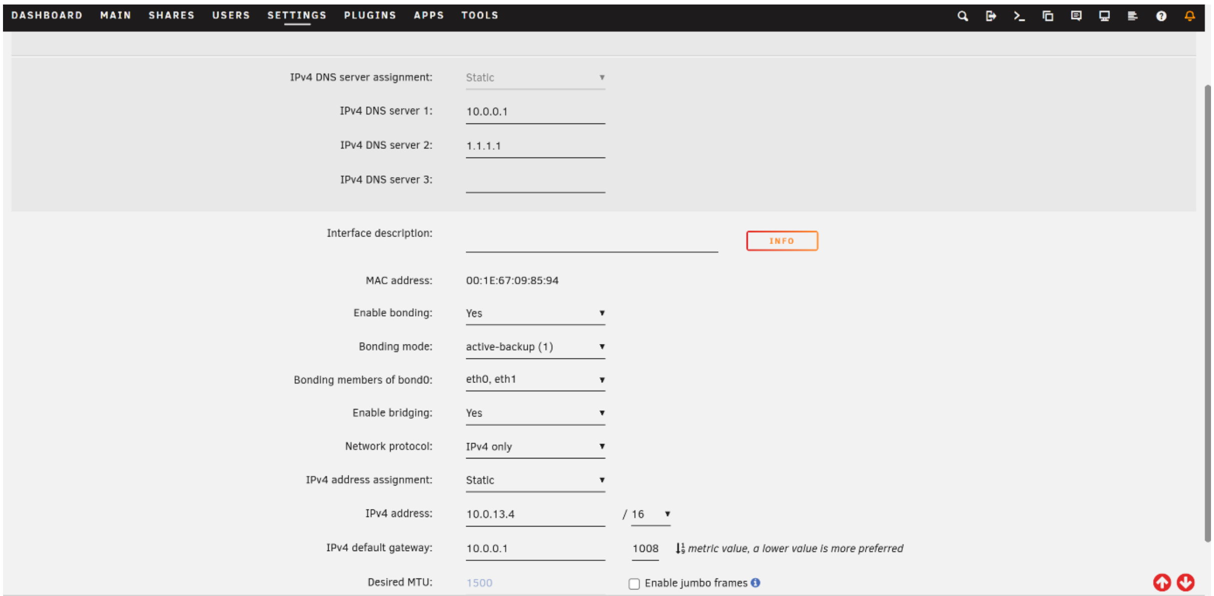


Figure 17: UnRAID network settings with the static address `10.0.13.4/16`, gateway, and DNS servers.

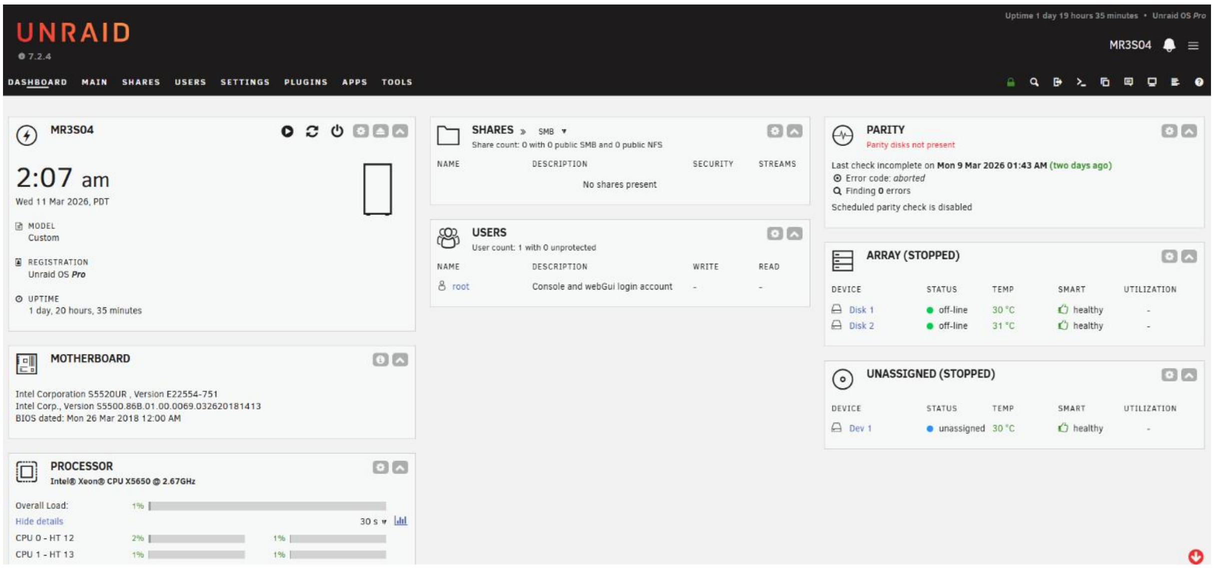


Figure 18: UnRAID dashboard after initial configuration, confirming that the system is reachable through the web interface.

5.3.3 OS Version Note

The installed version at the start of the session was **7.1.2**; version **7.2.4** was available. The update was initially postponed because a parity check was running, and rebooting would have interrupted it. After the parity check completed, we started the update process. The assignment requires documenting the latest available release, but does not require completing the upgrade during Part 4.

5.4 Web Interface Recovery

After clearing the config folder, the UnRAID web interface did not start automatically:

- localhost was unreachable
- Starting nginx returned **HTTP 502**
- PHP rendered only a blank page

Manual recovery commands:

```
cd /etc/rc.d
./rc.nginx start
./rc.php-fpm start
./rc.udev force-restart
/usr/local/sbin/emhttp &
```

Permanent fix – /boot/config/go:

To ensure the web interface starts automatically on every future boot, the file /boot/config/go was created with the following content:

```
#!/bin/bash
/usr/local/sbin/emhttp &
```

This file is executed by UnRAID at each startup. Adding emhttp to it guarantees the web management daemon launches automatically without manual intervention.

6 Part 5 – Rack Installation

6.1 Rack Plan

Before mounting, the installation was planned to ensure correct positioning, rail compatibility, and clean cable routing.

Parameter	Value
Rack	MR3S04
Rack size	24U
Server size	2U
Assigned rack position	5th server slot from top (U16–U15)

6.2 Rails and Mounting

Steps performed:

1. Identified the left (L) and right (R) rails by their markings.
2. Installed both rails in the correct orientation in the rack.
3. Verified that the rails locked correctly and slides moved smoothly.
4. Placed the server onto the rails, slid it into position, and secured it in the rack.

Verification	Result
Rails locked correctly	✓ Yes
Slides move smoothly	✓ Yes
Server seated and secured	✓ Yes

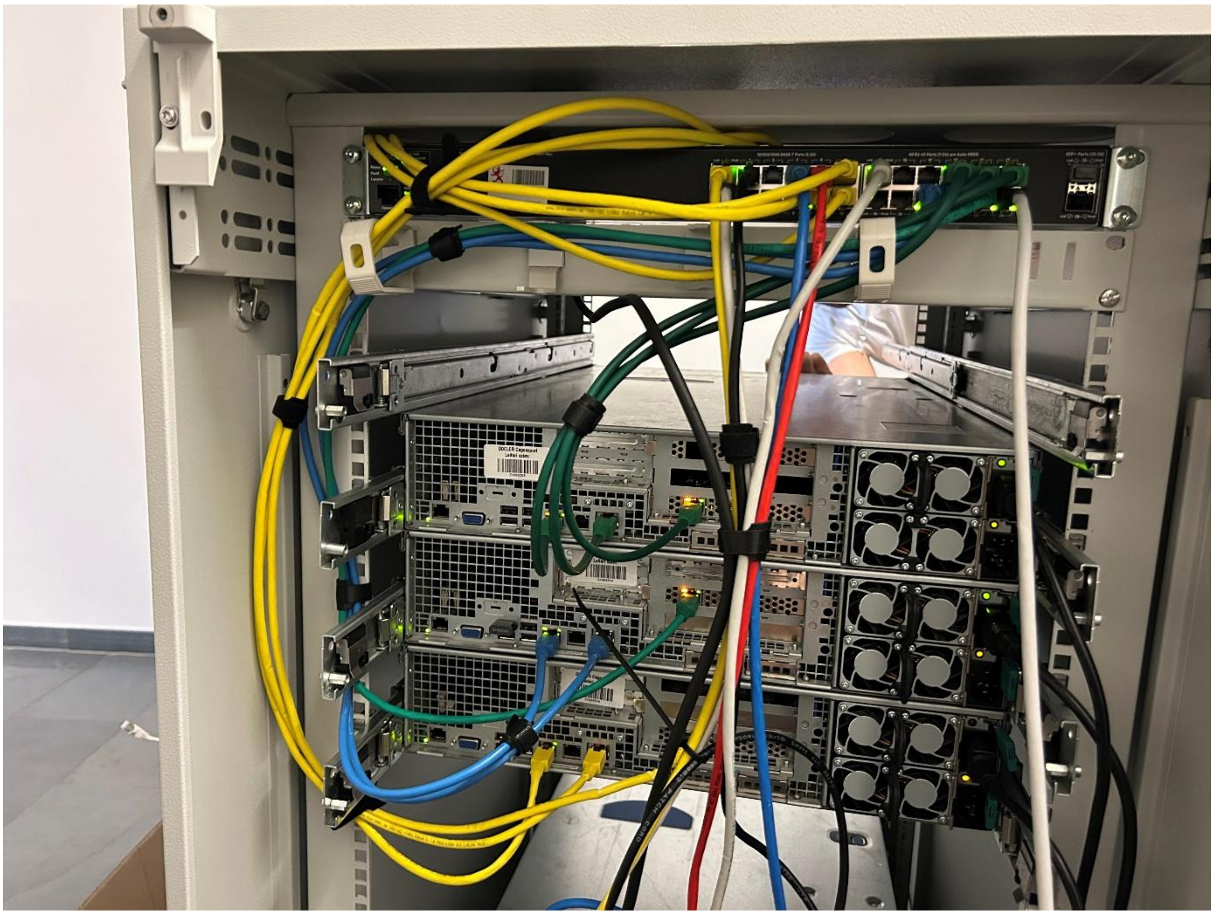


Figure 19: Server mounted in the rack with rails installed and cabling routed after physical installation.

6.3 Cabling

After mounting, the following connections were attached and cable management was completed:

Connection	Status
Power → PSU 1	✓ Connected (1 of 2 PSUs powered in rack — sufficient for operation)
LAN → RAID controller (Areca ARC-1880)	✓ Connected
LAN → Mainboard (primary NIC)	✓ Connected
LAN → RMM (Intel RMM3)	✓ Connected

Note: During staging, both PSUs were connected and the failover was tested (see §3.3). After rack installation, only one power cable was connected. The server operates normally on a single PSU.

Rear ports and LAN connections:

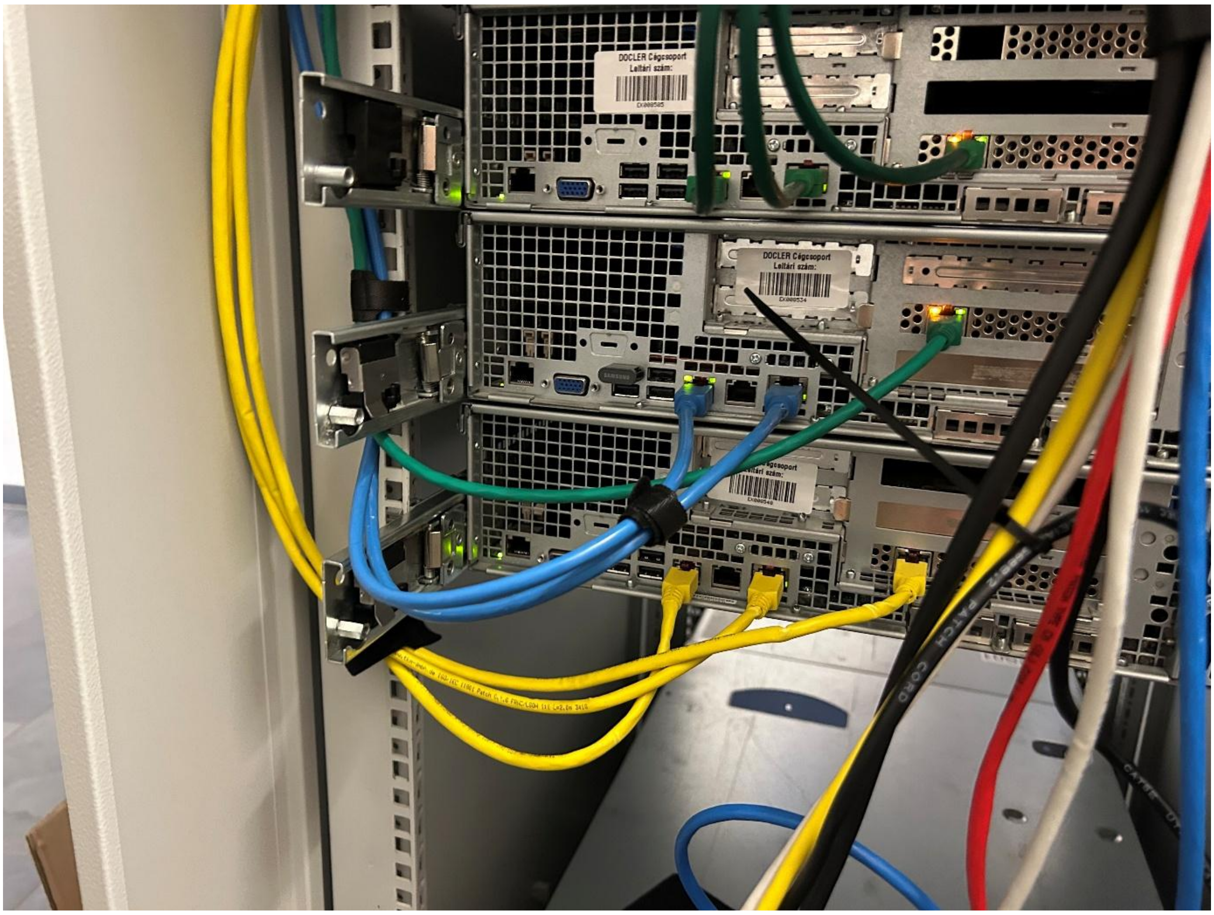


Figure 20: Rear panel cabling showing the connected mainboard LAN, RAID management LAN, and RMM port.

6.4 Access Verification

After cabling, remote management access was verified through the Intel RMM3 web interface:

Parameter	Value
RMM access URL	http://192.168.0.42
RMM model	Intel Remote Management Module 3 (RMM3)
Host Power Status	Host is currently ON
BMC available	Yes
BMC firmware revision	0.64
Management Engine (ME) firmware	01.12

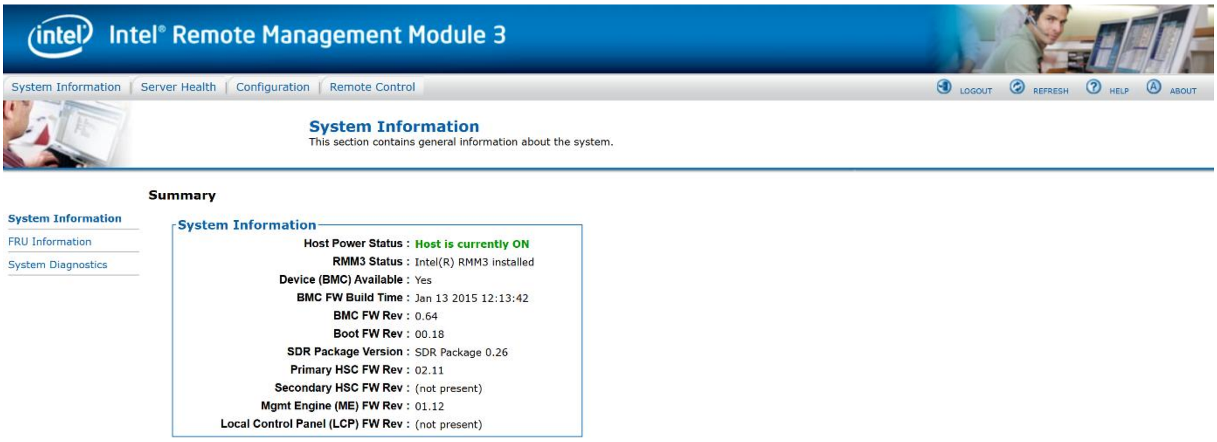


Figure 21: Intel RMM3 web interface confirming remote management access and current host power status.

7 Part 6 – NAS Configuration & Testing

7.1 RAID Configuration

7.1.1 Hardware RAID 5 – areca_raid5

The Areca ARC-1880 controller presents all three SAS disks as a single logical drive to the operating system. UnRAID has no knowledge of the underlying RAID structure – all redundancy is managed entirely by the controller hardware.

Parameter	Value
Volume name	`areca_raid5`
UnRAID device node	`sdb`
Identification	`areca_raid5_41aed13916860387_2001b4d2061683078`
Filesystem	XFS
Total size	280 GB
Used	5.39 GB
Free	274 GB
Temperature	30 °C
Errors	0

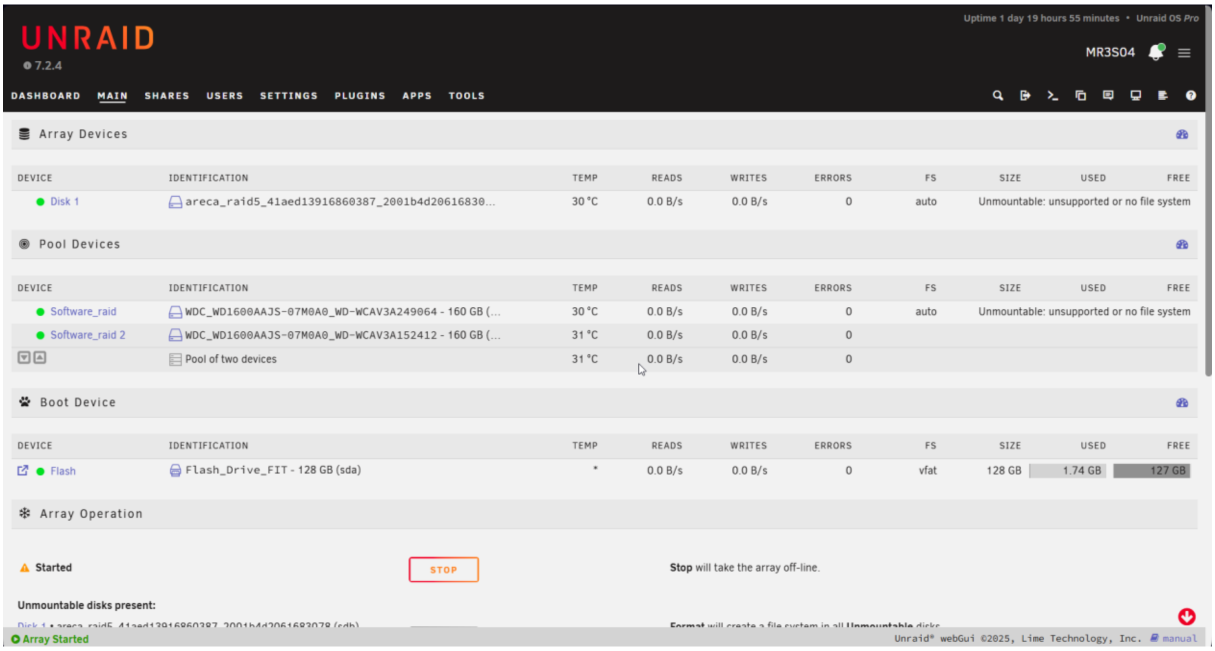


Figure 22: UnRAID array overview showing the started array with the hardware RAID 5 disk and software RAID 1 pool detected.

7.1.2 Software RAID 1 – software_raid1

A Btrfs RAID 1 pool was created using both native SATA disks. Btrfs RAID 1 mirrors all data across both disks, so a single disk failure results in no data loss.

Device	Identification	Node	FS	Size	Temperature
Software_raid	WDC_WD1600AAJS-07M0A0_WD-WCAV3A249064	sdc	Btrfs	160 GB	30 °C
Software_raid 2	WDC_WD1600AAJS-07M0A0_WD-WCAV3A152412	sdd	Btrfs	160 GB	31 °C

Issue encountered and resolved: After the initial pool configuration, UnRAID refused to start the array with the error "*Too many wrong and/or missing disks!*". Resolution: **Tools → New Config → Preserve current assignments: All**, then reassigned Parity to unassigned and both SATA disks to the `software_raid1` pool. Both disks initially showed as *Unmountable: unsupported or no file system* (expected for unformatted disks). After formatting via **Array Operation → Format**, the pool mounted correctly with a valid Btrfs filesystem.

7.2 File Sharing Scenario

The NAS was configured for a **family sharing scenario** with three user accounts and four shared folders distributed across both RAID arrays.

7.2.1 Shared Folders

Share	SMB	NFS	Storage pool	Available
Photos	Private	Disabled	areca_raid5 (RAID 5)	274 GB
Media	Private	Disabled	areca_raid5 (RAID 5)	274 GB
Documents	Private	Disabled	areca_raid5 (RAID 5)	274 GB
Backup	Private	Disabled	software_raid1 (RAID 1)	159 GB

Design decision: The `Backup` share is placed on `software_raid1` because RAID 1 mirroring guarantees a full copy on each disk ideal for backup data where a complete local copy matters most. The other shares use the larger `areca_raid5` array for bulk storage capacity.

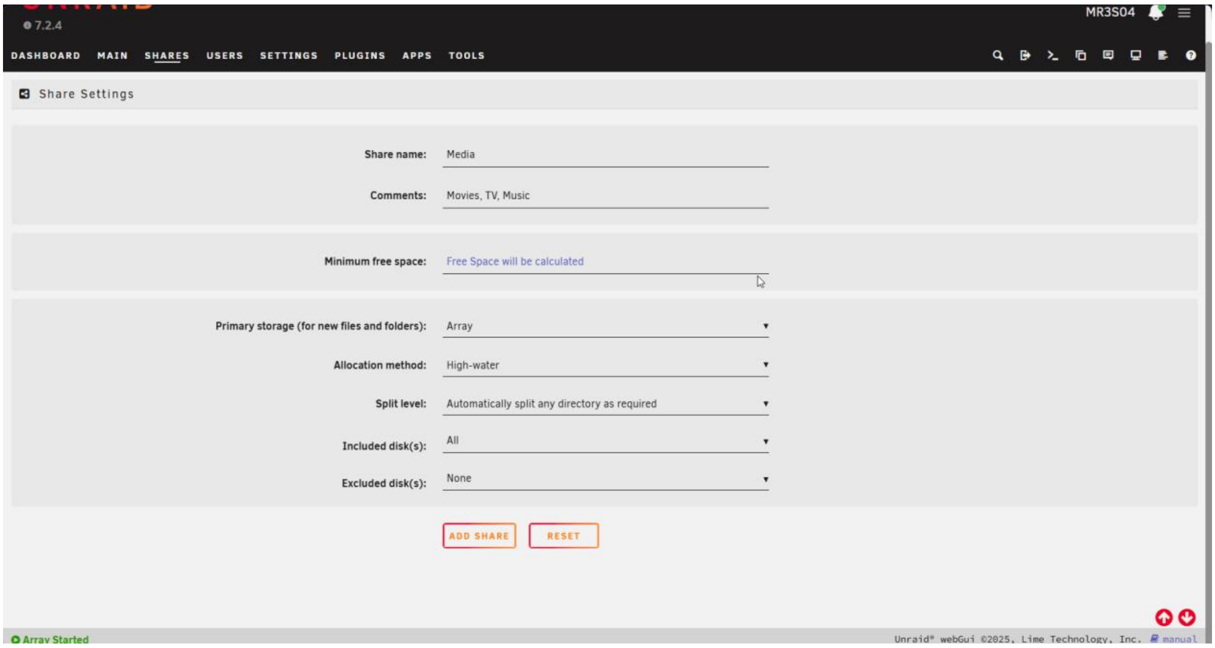


Figure 23: UnRAID share settings showing SMB access configured as Private for the family file-sharing scenario.

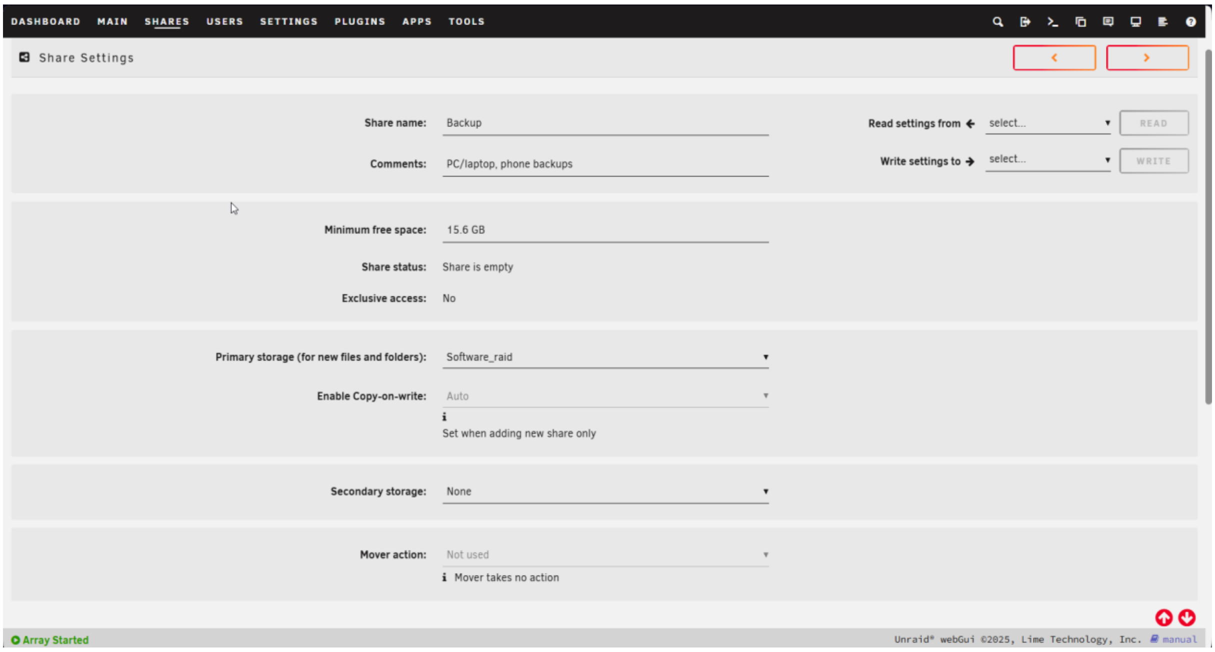


Figure 24: Backup share permission settings in UnRAID, used to restrict access for the child account.

7.2.2 Users and Permissions

User	Role	Photos	Media	Documents	Backup
gui	Parent 1	Read/Write	Read/Write	Read/Write	Read/Write
andrea	Parent 2	Read/Write	Read/Write	Read/Write	Read/Write
marios	Child 1	Read only	Read only	Read only	No access

7.2.3 Multi-Platform Access

All shares are accessible via **SMB**, which is natively supported on all three major desktop platforms:

Platform	Protocol	Access method
Windows	SMB	File Explorer → \\MR3S04
Linux	SMB	Nautilus / smbclient \\MR3S04\<share>
macOS	SMB	Finder → Go → Connect to Server → smb://MR3S04

7.3 Tests Performed

7.3.1 Upload / Download Test

File	Size	Share	User	Protocol	Result
test.txt	24 bytes	\\MR3S04\Documents	gui	SMB	✓ Success
IMG_RAM.jpeg	2.22 MB	\\MR3S04\Media	gui	SMB	✓ Success

Client OS: Windows (File Explorer).

7.3.2 Permission Test

User	Share tested	Expected access	Observed result
gui (Parent 1)	Backup	Read/Write	✓ Full access confirmed
andrea (Parent 2)	Backup	Read/Write	✓ Full access confirmed
marios (Child 1)	Backup	No access	✓ Access denied as expected

The permission differences were confirmed: the child account was correctly blocked from the Backup share while both parent accounts had full read/write access.

7.4 iSCSI Research

7.4.1 What is iSCSI?

iSCSI (Internet Small Computer Systems Interface) is a block-level storage protocol that transports SCSI commands over a standard TCP/IP network. From the client's perspective, the remote NAS storage appears as a **local block device** — a raw disk that the client itself formats and manages.

Term	Definition
Target	The server side (NAS) that exposes block storage
Initiator	The client side that connects to and uses the storage
LUN	Logical Unit Number – the specific block device exposed by the target
IQN	Unique string identifier for an iSCSI initiator or target

Typical use cases: virtualisation platforms (VMware, Hyper-V, Proxmox), databases, and applications that require raw local disk access rather than a network file share.

7.4.2 iSCSI vs NFS vs SMB

Protocol	Level	Appears on client as	Best use case
iSCSI	Block	Local disk (e.g. Disk 2 in Windows Disk Management)	VM storage, databases, apps requiring raw block access
NFS	File	Mounted folder (e.g. /mnt/nas)	Linux/Unix file sharing, servers, containers
SMB	File	Network drive (e.g. \\SERVER\Share)	Windows/mixed-OS home and office file sharing

The fundamental difference: iSCSI exports **raw blocks** and the **client** manages its own filesystem. NFS and SMB export **files and folders** and the **NAS** manages the filesystem.

7.4.3 iSCSI Support in UnRAID

UnRAID does not include a built-in iSCSI target in its core OS. iSCSI support is provided through the Community Apps ecosystem:

Plugin	Author	Function
Unraid iSCSI	SimonFair	iSCSI target – allows UnRAID to expose LUNs to initiators
ich777/iscsi-initiator	ich777	iSCSI initiator – allows UnRAID itself to connect to iSCSI targets

7.5 iSCSI Test Implementation

A practical test was carried out using the UnRAID NAS as the **iSCSI target** and a Windows PC as the **initiator**.

Step 1 – Install the iSCSI Plugin

1. Open the **Apps** tab (Community Applications) in the UnRAID web UI.
2. Search for **"Unraid iSCSI"** (by SimonFair).
3. Install with default settings.

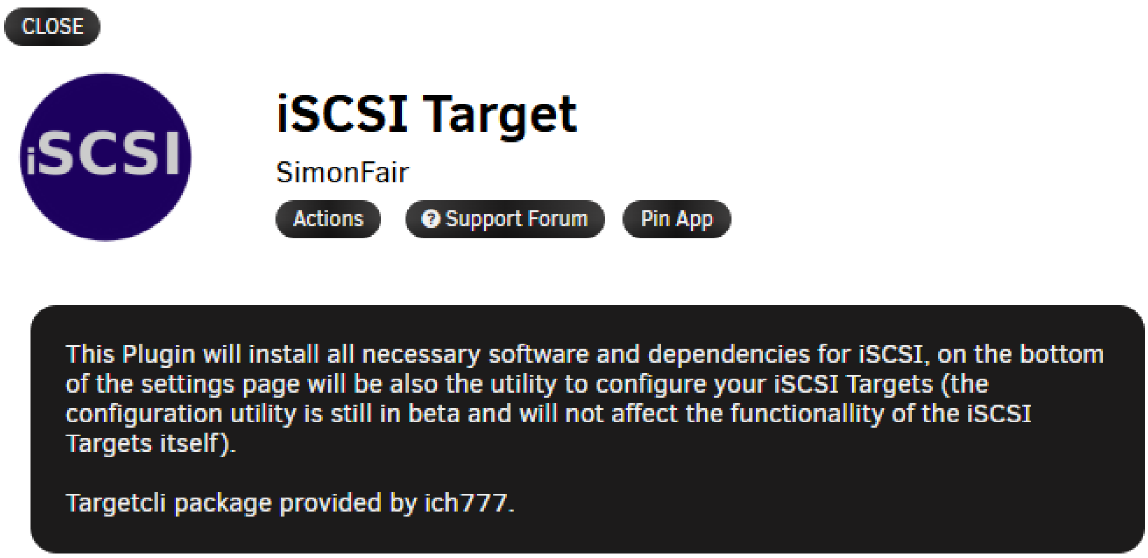


Figure 25: Community Applications page showing the Unraid iSCSI plugin installed on the NAS.

Step 2 – Create a LUN and Target

1. In the iSCSI plugin GUI, create a **file-based LUN** stored on the areca_raid5 hardware RAID 5 array (so the LUN benefits from hardware RAID protection).
2. Create a new **iSCSI target** and attach the LUN as LUN 0.

Evidence: iSCSI-setup1.png, iSCSI-setup2.png, iSCSI-setup3.png, iSCSI-setup4.png

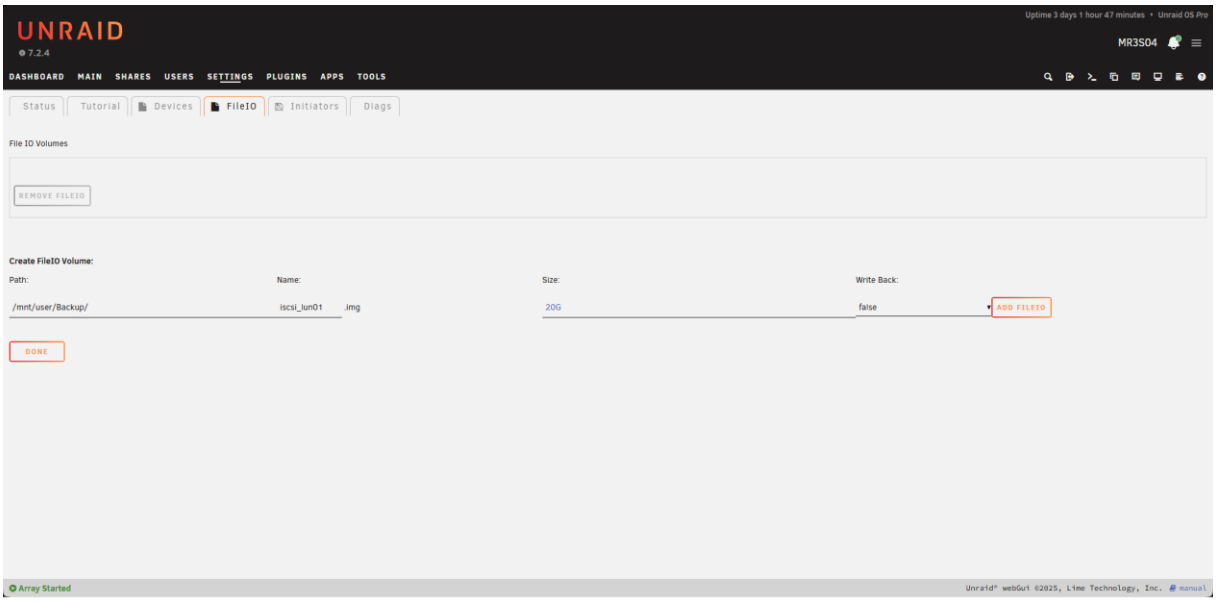


Figure 26: iSCSI plugin screen used to create the file-based LUN on the `areca_raid5` array.

Figure 27: iSCSI target configuration with the new target created and the LUN attached as `LUN 0`.

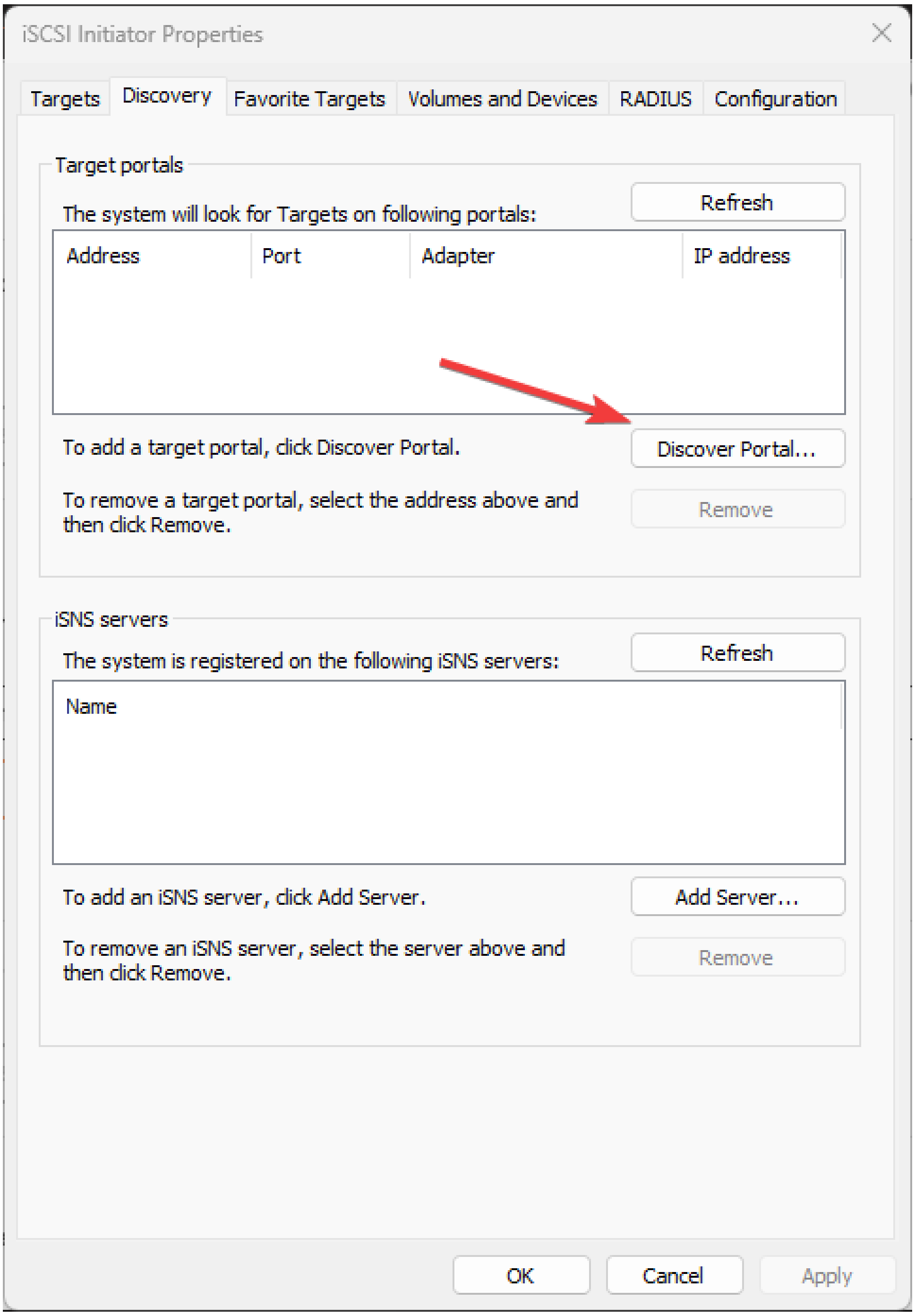


Figure 28: iSCSI initiator ACL configuration showing the authorized Windows initiator IQN.

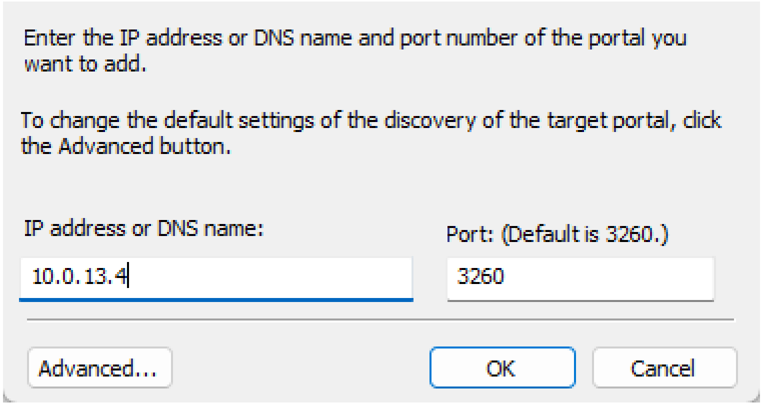


Figure 29: iSCSI target for IP connection

Step 3 – Retrieve the Windows Initiator Name

1. On the Windows PC, open **iSCSI Initiator** (search in Start menu).
2. Navigate to the **Configuration** tab.
3. Copy the **Initiator Name** (format: iqn.1991-05.com.microsoft:<hostname>).

Figure 30: Windows iSCSI Initiator configuration tab showing the initiator IQN copied for authorization.

Step 4 – Authorise the Initiator on UnRAID

1. In UnRAID → **Settings** → **iSCSI**, locate the **Initiators** section.
2. Click **Add Initiator** and paste the Windows IQN.
3. Confirm the Windows IQN appears in the **ACL** of the target.

***Important:** Without this step, the Windows initiator receives an authorisation failure when connecting. The plugin only supports Add and Remove (no editing), so the IQN must be entered correctly from the start.*

Step 5 – Connect from Windows and Verify

1. In **iSCSI Initiator** on Windows, enter the UnRAID server IP in the **Targets** tab and click **Quick Connect**.
2. The target status changes from *Inactive* to **Connected**.
3. Open **Disk Management** — the iSCSI LUN appears as a new disk, just like a local drive.
4. Initialise the disk, create a new simple volume, assign a drive letter, and format as NTFS.
5. Copy test files to the volume to verify end-to-end functionality.

The disk behaves exactly like a local disk from the user's perspective — but all data is stored on the UnRAID server over the network, and the underlying LUN is protected by the hardware RAID 5 array.

Figure 31: Windows iSCSI Initiator

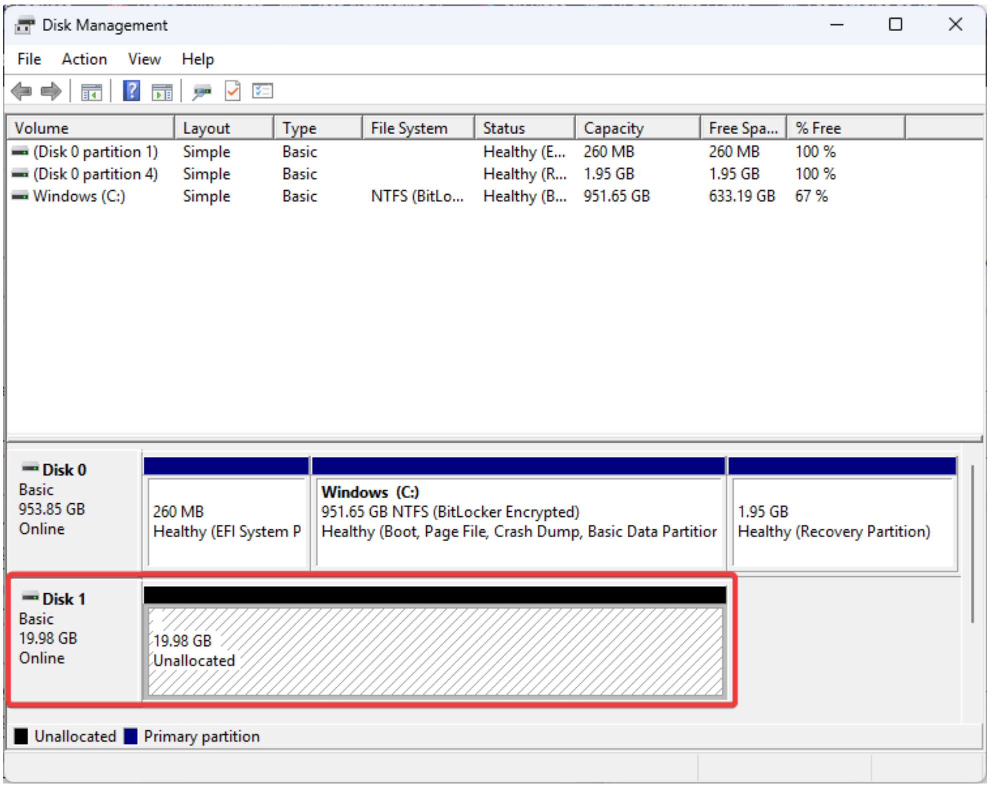


Figure 32: Windows Disk Management detecting the newly attached iSCSI disk before formatting.

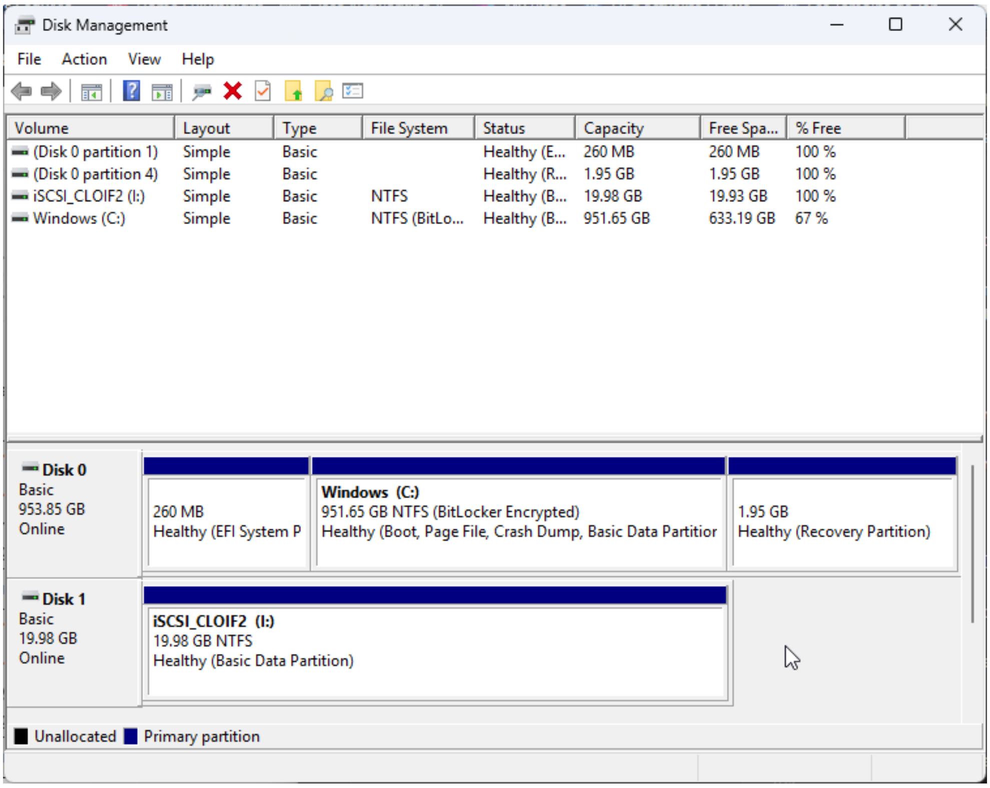


Figure 33: Windows Disk Management after initializing and formatting the iSCSI disk as a usable volume.

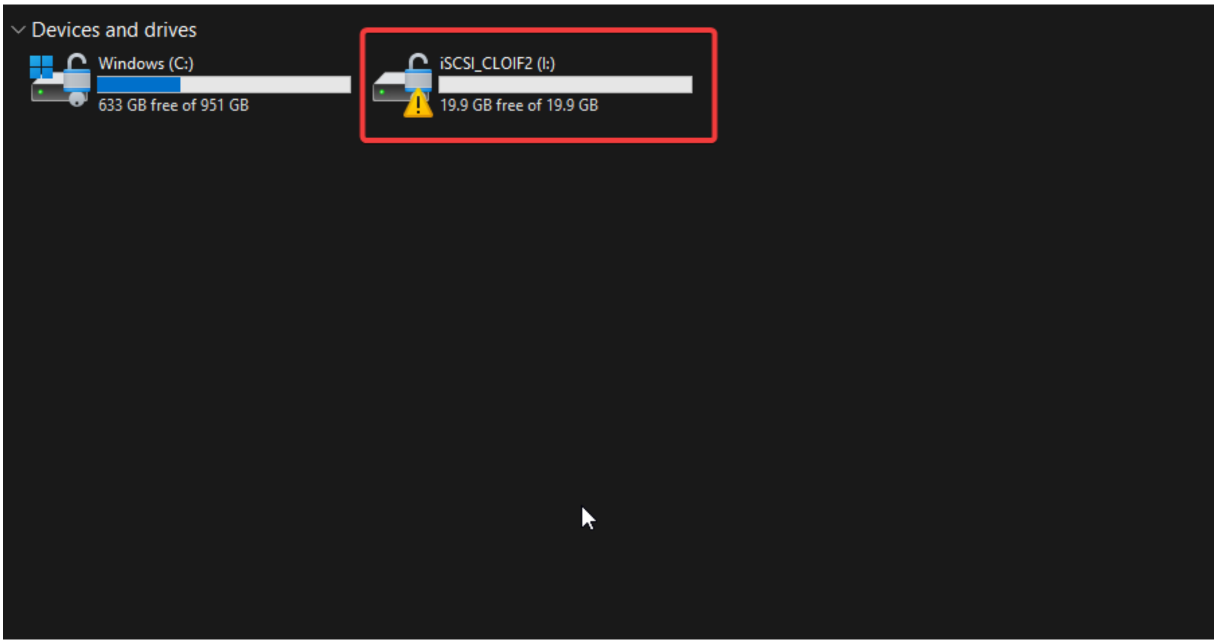


Figure 34: Test files copied to the mounted iSCSI volume, confirming end-to-end block storage access.

8 Network Architecture

8.1 IP Address Summary

Interface	IP address	Subnet mask	Gateway	DNS
UnRAID LAN (mainboard NIC)	`10.0.13.4`	`255.255.0.0 (/16)`	`10.0.0.1`	`10.0.0.1` · `1.1.1.1`
RMM3 (Intel remote management)	`192.168.0.42`	`255.255.255.0 (/24)`	`192.168.0.254`	`8.8.8.8`
RAID controller (Areca ARC-1880)	`192.168.0.142`	`255.255.255.0 (/24)`	`192.168.0.254`	`8.8.8.8`

Parameter	Value
Hostname	`MR3S04`
UnRAID web UI	http://10.0.13.4
RAID web UI	http://192.168.0.142
RMM web UI	http://192.168.0.42

8.2 Infrastructure Diagrams

Network topology:

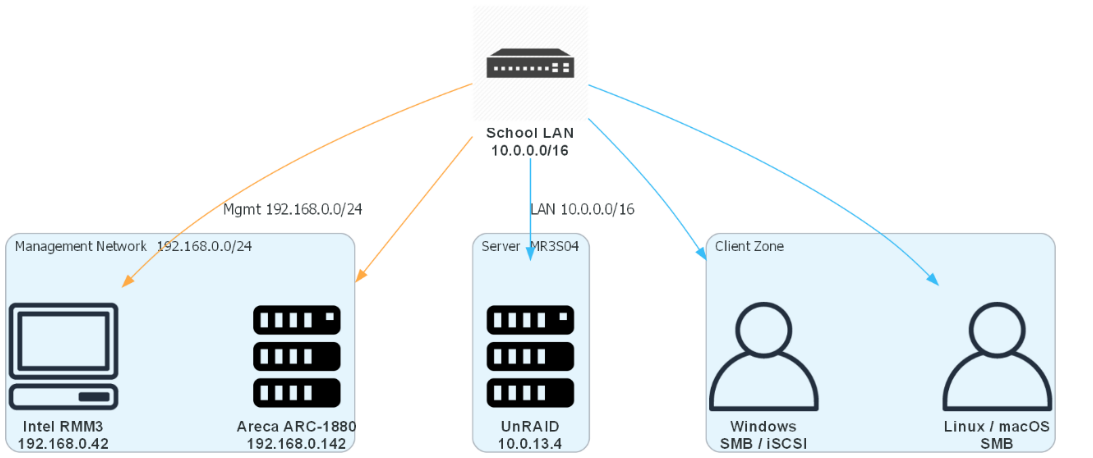


Figure 35: Network topology of server `MR3S04`, showing the UnRAID data interface, RMM management link, and Areca management link.

Storage layout:

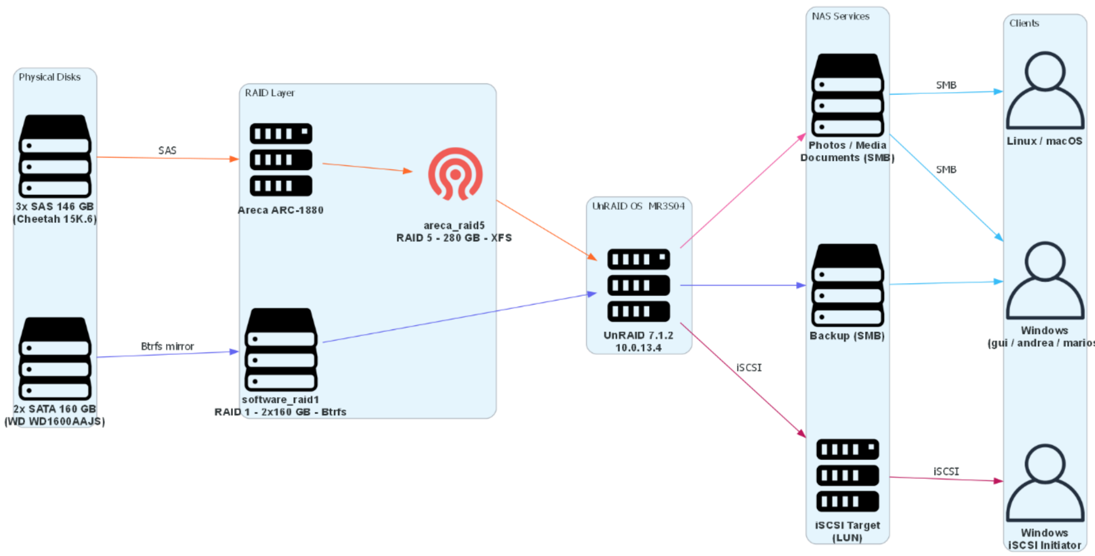


Figure 36: Storage layout showing the SATA RAID 1 pool, the Areca hardware RAID 5 array, the main shares, and the iSCSI LUN placement.

9 Sources

#	Source	URL	Accessed
1	UnRAID official website	https://unraid.net	2026-03
2	UnRAID documentation	https://docs.unraid.net	2026-03
3	Areca ARC-1880 product page	https://www.areca.com.tw	2026-03
4	Intel SR2600URBRPR product page	https://ark.intel.com	2026-03
5	iSCSI overview (Wikipedia)	https://en.wikipedia.org/wiki/ISCSI	2026-03
6	Unraid iSCSI plugin by SimonFair	https://github.com/SimonFair/unraid-iscsi	2026-03
7	Btrfs RAID documentation	https://btrfs.readthedocs.io	2026-03
8	Intel SR2600UR/SR2625UR Service Guide (E51243-006)	Intel document (local PDF)	2026-03

Glossary

Term	Definition
ACL	Access Control List – a set of rules defining which users or systems can access a resource
Backplane	Internal circuit board connecting drive bays to the server mainboard or storage controller
BIOS	Basic Input/Output System – firmware that initialises hardware before the operating system loads
BMC	Baseboard Management Controller – dedicated chip enabling out-of-band server management
Btrfs	B-tree Filesystem – a modern Linux filesystem with built-in RAID, checksumming, and snapshot support
DAS	Direct Attached Storage – storage connected directly to a single computer
DDR3	Double Data Rate 3 – third generation of synchronous DRAM memory
DHCP	Dynamic Host Configuration Protocol – automatic IP address assignment over a network
DNS	Domain Name System – translates hostnames into IP addresses
Docker	Container platform for running isolated, lightweight application environments
EFI / UEFI	(Unified) Extensible Firmware Interface – modern replacement for BIOS
Gateway	Router that connects a local network to other networks or the internet
HDD	Hard Disk Drive – mechanical spinning-disk storage device
IQN	iSCSI Qualified Name – unique string identifier for an iSCSI initiator or target
iSCSI	Internet Small Computer Systems Interface – protocol that sends SCSI commands over TCP/IP for block-level storage

LAN	Local Area Network – a network connecting devices within a limited area
LUN	Logical Unit Number – a logical identifier for a block storage device exposed by an iSCSI target
NAS	Network Attached Storage – file storage shared over a network
NFS	Network File System – Unix/Linux file-sharing protocol
NTFS	New Technology File System – standard Windows filesystem
OS	Operating System
PCIe	Peripheral Component Interconnect Express – high-speed serial expansion bus
RAID	Redundant Array of Independent Disks – combines multiple disks for redundancy or performance
RAID 1	Disk mirroring – data is written identically to two or more disks
RAID 5	Striping with distributed parity – requires ≥ 3 disks, survives one disk failure
RMM	Remote Management Module – hardware module enabling out-of-band server access (KVM, power control)
SAN	Storage Area Network – high-speed block storage network used in enterprise data centres
SAS	Serial Attached SCSI – high-performance enterprise disk interface
SATA	Serial ATA – common disk and SSD interface for consumer and server hardware
SMB	Server Message Block – Windows file-sharing protocol, also supported natively on Linux and macOS
SSH	Secure Shell – encrypted remote command-line protocol
TCP/IP	Transmission Control Protocol / Internet Protocol – core networking protocol suite
USB	Universal Serial Bus
VGA	Video Graphics Array – analogue video output interface
VM	Virtual Machine
XFS	A high-performance 64-bit journalling filesystem, default storage format in many Linux distributions

Andrea Girotto | Guilherme Videira Marques

THANK YOU FOR READING